Integral Secant Distributions and Improved Bounds for Complete (k,n)-Arcs

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Abstract

We study complete (k, n)-arcs in finite projective planes through the numbers of n-secants incident with each point. We split the exact block-pair count between the arc and its complement and derive sharp integer bounds for both degree sequences. The real relaxation recovers the classical incidence bound. Integer secant distributions retain arithmetic information lost by spectral mixing and give a linear obstruction at every rational equality parameter. For each factorization $\lambda = uv$, we obtain an explicit lower bound for (k, n)-arcs such that every external point lies on at least λ n-secants, where $q = (u + v + 1)m$ and $n = (u + 1)m + 1$. This bound improves the spectral coefficient by a positive multiple of q. On characteristic-compatible families, near equality forces bounded modular repair, and completeness converts that defect into further linear improvements. In the ordinary-completeness branch over $\mathrm{PG}(2, 3^h)$, this proves

$$k \geq \frac{q^2}{3} + \frac{4q}{3} - o(q), \qquad n = \frac{2q}{3} + 1.$$

Over characteristic two, an analogous parity argument gives a further linear improvement when every external point lies on two n-secants. The integer incidence theorem also applies to selected blocks of any symmetric design and yields formulations for minimal multiple blocking sets and nonextendible projective codes.

Keywords. complete (k, n)-arc; n-secant; multiple blocking set; symmetric design; projective code; incidence bound; modular stability.

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1 Introduction

A (k, n)-arc in a projective plane is a set of k points meeting every line in at most n points, with some line meeting it in exactly n points. It is complete if it is maximal under inclusion, or equivalently if every external point lies on an n-secant. The minimum size of a complete (k, n)-arc is denoted by $t_n(2, q)$ in $\mathrm{PG}(2, q)$. Lower bounds for this quantity are usually obtained by counting n-secants or, dually, by bounding minimal multiple blocking sets; see Alabdullah–Hirschfeld [1] and Bishnoi–Mattheus–Schillewaert [7]. Recent work has also developed algebraic and curve-based methods for proving completeness itself; see, for example, Bastioni–Micheli [5], Korchmáros–Nagy–Szőnyi [10], and Bartoli–Timpanella [4].

The incidence estimate used in the existing lower bounds is an instance of the standard block-design incidence inequality; we use the normalization of Lund–Saraf [12] and the variance formulation of Murphy–Petridis [13]. Exact quotient equations for tactical decompositions are classical [6], but the selected family of all n-secants need not be tactical. Ramani's recent first-incidence identity for uniform intersecting linear hypergraphs

[14] provides a related degree-defect viewpoint. Our argument uses both point classes and the intervals determined by their exact pair-count extrema.

This paper keeps the n-secants as a family rather than eliminating their number at the first counting step. The degrees of that family on the arc and on its complement have fixed sums, and their pair counts add to the number of pairs of secants. Integer convexity gives the exact extrema of each pair count. The two ranges must overlap. Replacing the integer extrema by quadratic real averages recovers the classical incidence-mixing inequality; retaining them gives the new bounds.

The first application concerns ordinary completeness in characteristic three.

Theorem 1.1 (Characteristic-three bound). *For every $\varepsilon > 0$, there is h_0 such that, for $h \geq h_0$, every complete $(k, 2q/3+1)$-arc in* $\mathrm{PG}(2,q)$, *where $q = 3^h$, satisfies*

$$k \geq \frac{q^2}{3} + \left(\frac{4}{3} - \varepsilon\right) q. \tag{1.1}$$

Equivalently,

$$t_{2q/3+1}(2,q) \geq \frac{q^2}{3} + \frac{4q}{3} - o(q) \qquad (q = 3^h \to \infty).$$

The incidence-mixing bound gives only $q^2/3 + q + O(1)$ in this regime. The integer degree bounds first raise the linear coefficient to $6/5$; a modular repair theorem of Szőnyi–Weiner [16] then raises it to $4/3$. The second step uses more than modular stability: each point changed in the dual repaired multiset becomes a primal line on which completeness forces a linear amount of external secant excess.

The arithmetic source of the first improvement is visible whenever the real incidence bound is tight at a rational parameter. Fix a coverage multiplicity λ, factor it as $\lambda = uv$, and put $d = u + v + 1$. For $h \in \mathbb{Z}$, define

$$c_{\mathrm{bal}} = \frac{u(uv^2 + 4uv + 2u + 2v^2 + 4v + 1)}{2uv + u + v}, \tag{1.2}$$

$$h_* = -\frac{u(v+1)(u^2v + u^2 - v)}{2uv + u + v}, \tag{1.3}$$

$$\mu = \frac{2v}{(u+v)(2uv + u + v + 1)},$$

$$L(h) = d - \frac{h}{u}, \qquad C(h) = c_{\mathrm{bal}} + \mu(h - h_*),$$

$$c_{\mathbb{Z}} = \min_{h \in \mathbb{Z}} \max\{L(h), C(h)\}. \tag{1.4}$$

Theorem 1.2 (Rational equality families). *Let $u, v \geq 1$, $\lambda = uv$, and $d = u + v + 1$. Suppose that a $(k, (u+1)m+1)$-arc in a projective plane of order $q = dm$ has at least λ n-secants through every external point. Then*

$$k \geq udm^2 + c_{\mathbb{Z}} m - O_{u,v}(1). \tag{1.5}$$

Moreover, the classical incidence-mixing bound has linear coefficient

$$c_{\mathrm{mix}} = \frac{u(uv + 3u + 2v + 3)}{2u + 1},$$

and

$$c_{\mathrm{bal}} - c_{\mathrm{mix}} = \frac{u(v+1)(u^2 + u + 1)}{(2u+1)(2uv + u + v)} > 0. \tag{1.6}$$

Every rational equality parameter of the continuous bound whose limiting internal degree is an integer $a > \lambda$ arises from exactly one ordered factorization $\lambda = uv$.

Thus the strict improvement is not an isolated rounding effect. At fixed coverage multiplicity there is one equality family for each divisor, and every family has a positive linear integrality penalty. The two integers adjacent to h_* determine $c_{\mathbb{Z}}$, so the coefficient is explicit.

The same integer incidence condition holds in arbitrary symmetric designs. In a projective plane there are two further bounds. At most $\lfloor (k-1)/(n-1) \rfloor$ n-secants pass through a point of the arc, and at most $\lfloor k/n \rfloor$ pass through an external point. These geometric bounds sharpen the general condition for complete arcs.

In characteristic two the first family in which the two first-order bounds are simultaneously tight yields a second application.

Theorem 1.3 (Characteristic-two parity bound)**.** *For every $\varepsilon > 0$, there is h_0 such that, for $h \geq h_0$, if $q = 2^h$ and every external point of a $(k, 3q/4+1)$-arc in* $\mathrm{PG}(2, q)$ *lies on at least two $(3q/4+1)$-secants, then*

$$k \geq \frac{q^2}{2} + \left(\frac{73}{48} - \varepsilon\right) q. \tag{1.7}$$

The proof separates odd and even repair support. The parity of the selected secant family fixes the parity of the support, while the symmetric difference of its generator lines supplies respectively $q/4 - O(1)$ or $q/2 - O(1)$ external points of excess degree.

The projective-code translation is immediate. A spanning point set in $\mathrm{PG}(r-1, q)$ gives a projective $[k, r, k-n]_q$ code. Hyperplanes meeting the point set in n columns correspond to minimum-weight codewords, and a candidate extension column is obstructed by each such hyperplane through it. The same integer degree theorem therefore bounds nonextendibility with multiple minimum-weight witnesses. We keep the finite-geometry language primary and record this translation after the planar applications.

The paper is organized as follows. Section 2 proves the symmetric-design inequality and identifies incidence mixing as its real relaxation. The next two sections specialize to n-secants and prove Theorem 1.2. The modular-stability sections prove Theorems 1.1 and 1.3. The final section gives the blocking-set and coding translations and records the exact computational checks used to guard the finite algebra.

2 Sharp integer degree bounds

For integers $N \geq 1$ and $R \geq 0$, write $R = aN + w$ with $0 \leq w < N$, and define

$$\Phi_{\min}(N, R) = (N - w)\binom{a}{2} + w\binom{a+1}{2}. \tag{2.1}$$

If $L \leq U$, $NL \leq R \leq NU$, put $E = R - NL = t(U - L) + w$, where $0 \leq w < U - L$, and define

$$\Phi_{\max}(N, L, U, R) = N\binom{L}{2} + LE + t\binom{U-L}{2} + \binom{w}{2}. \tag{2.2}$$

When $L = U$, the latter expression has its evident constant interpretation.

We will use the following exact form of the balancing loss. If $R = Na + w$, then every integer sequence with sum R satisfies

$$\sum_{i=1}^{N}\binom{z_i}{2} - \Phi_{\min}(N, R) = \frac{1}{2}\sum_{i=1}^{N}(z_i - a)(z_i - a - 1). \tag{2.3}$$

Every summand on the right is nonnegative, and every index with $z_i \notin \{a, a+1\}$ contributes at least one to the left side. Thus an $O(m)$ total balancing loss permits only $O(m)$ departures from the two adjacent balanced degrees. This quantitative consequence, rather than an unstated stability principle, will control the exceptional lines in Sections 5 and 6.

Lemma 2.1 (Balanced and filled degree sequences)**.** *Among integer sequences $z_1, \dots, z_N$ with sum R, the minimum of $\sum_i \binom{z_i}{2}$ is $\Phi_{\min}(N, R)$. With the additional bounds $L \le z_i \le U$, its maximum is $\Phi_{\max}(N, L, U, R)$.*

Proof. If $z_i \ge z_j + 2$, replacing (z_i, z_j) by $(z_i - 1, z_j + 1)$ lowers the sum by $z_i - z_j - 1$. Repetition leaves only the adjacent values $a, a+1$, proving (2.1). For the maximum, the reverse exchange moves one unit from a smaller nonminimal coordinate to a larger nonmaximal one and raises the sum. Hence all but at most one coordinate lie at L or U; counting the filled coordinates gives (2.2). □

Theorem 2.2 (Exact integer incidence condition)**.** *Let $\mathcal{D}$ be a symmetric 2-(v, K, Λ) design whose point set is partitioned into $A \sqcup B$. Let $\mathcal{S}$ be a family of T blocks, each meeting A in s points. Suppose the selected-block degrees satisfy integer bounds*

$$L_x \le z_x \le U_x \qquad (x \in A \sqcup B).$$

At the fixed totals

$$\sum_{x \in A} z_x = sT, \qquad \sum_{x \in B} z_x = (K - s)T,$$

let $[P_A^-, P_A^+]$ and $[P_B^-, P_B^+]$ be the exact minimum and maximum of the two sums $\sum \binom{z_x}{2}$ over their respective integer boxes. Then

$$[P_B^-, P_B^+] \cap \left[\Lambda \binom{T}{2} - P_A^+, \Lambda \binom{T}{2} - P_A^- \right] \ne \varnothing. \tag{2.4}$$

For uniform boxes the endpoints are given by Lemma 2.1; for heterogeneous boxes, repeated unit transfers toward the lower bounds give the minimum, and repeated transfers toward the upper bounds give the maximum.

Proof. Count triples (x, S_1, S_2) with $S_1, S_2 \in \mathcal{S}$, $S_1 \ne S_2$, and $x \in S_1 \cap S_2$. Two blocks of a symmetric design meet in exactly Λ points, so

$$\sum_{x \in A} \binom{z_x}{2} + \sum_{x \in B} \binom{z_x}{2} = \Lambda \binom{T}{2}. \tag{2.5}$$

Each sum lies in its exact integer interval. Reflecting the interval for A through the right side of (2.5) gives (2.4). □

The real relaxation is familiar, but the following formulation makes the loss of information exact.

Corollary 2.3 (Incidence mixing as the real relaxation)**.** *In the setting of Theorem 2.2, assume additionally that A and B are nonempty and that $T > 0$. Then*

$$T \left(\frac{s^2}{|A|} + \frac{(K - s)^2}{|B|} - \Lambda \right) \le K - \Lambda. \tag{2.6}$$

This is algebraically identical to the incidence-graph mixing inequality for the point set B and block family $\mathcal{S}$. Equality requires constant selected-block degree on both point classes.

Proof. Cauchy–Schwarz and (2.5) give

$$\frac{s^2T^2}{|A|} - sT + \frac{(K-s)^2T^2}{|B|} - (K-s)T \le \Lambda T(T-1).$$

Divide by T and use $K = s+(K-s)$. Conversely, expanding the standard mixing inequality and substituting $K(K-1) = \Lambda(v-1)$ gives the same expression. Equality in the two Cauchy inequalities is equivalent to constant degrees. □

For uniform lower bounds without an upper cap, the balancing remainder has the closed form

$$\Phi_{\min}(N,R) = \frac{R^2/N - R}{2} + \frac{w(N-w)}{2N}, \qquad R \equiv w \pmod N, \qquad 0 \le w < N. \tag{2.7}$$

The last term is precisely what the real relaxation discards. Along some rational equality families, it has linear size and cannot be absorbed into lower-order error.

3 The distribution of n-secants

Let $\mathcal{K}$ be a (k,n)-arc in a projective plane of order q. We use the standard secant notation

$$\tau_i = \#\{i\text{-secants of } \mathcal{K}\},$$

and write $\rho_i(P)$ for the number of i-secants through $P \in \mathcal{K}$, and $\sigma_i(Q)$ for the number through $Q \notin \mathcal{K}$. Only $i = n$ will occur below. Put

$$b = q^2 + q + 1 - k, \qquad \tau = q + 1 - n, \qquad T = \tau_n.$$

The geometry supplies two bounds that need not hold for a selected block family in an arbitrary design.

Lemma 3.1 (Secants in a pencil)**.** *For every $P \in \mathcal{K}$ and $Q \notin \mathcal{K}$,*

$$\rho_n(P) \le D := \left\lfloor \frac{k-1}{n-1} \right\rfloor, \qquad \sigma_n(Q) \le M := \left\lfloor \frac{k}{n} \right\rfloor. \tag{3.1}$$

Moreover,

$$\sum_{P\in\mathcal{K}} \rho_n(P) = nT, \qquad \sum_{Q\notin\mathcal{K}} \sigma_n(Q) = \tau T, \tag{3.2}$$

$$\sum_{P\in\mathcal{K}} \binom{\rho_n(P)}{2} + \sum_{Q\notin\mathcal{K}} \binom{\sigma_n(Q)}{2} = \binom{T}{2}. \tag{3.3}$$

Proof. The n-secants through P have disjoint sets of $n-1$ further points of $\mathcal{K}$. Those through Q have disjoint n-point intersections with $\mathcal{K}$. This proves (3.1). The first two equations count incidences. Every two distinct projective lines meet once, either in $\mathcal{K}$ or outside it, which gives (3.3). □

We will sometimes impose the stronger hypothesis

$$\sigma_n(Q) \ge \lambda \qquad (Q \notin \mathcal{K}), \tag{3.4}$$

where $\lambda \ge 1$. For $\lambda = 1$, this is precisely completeness. Rather than introduce another name for (3.4), we state the multiplicity explicitly in each result.

Corollary 3.2 (Integer feasibility for the n-secants)**.** *Assume* (3.4). *Then there is an integer T satisfying*

$$\left\lceil \frac{\lambda b}{\tau} \right\rceil \leq T \leq \left\lfloor \frac{k(k-1)}{n(n-1)} \right\rfloor, \tag{3.5}$$

$$[\Phi_{\min}(b, \tau T), \Phi_{\max}(b, \lambda, M, \tau T)] \cap [\binom{T}{2} - \Phi_{\max}(k, 0, D, nT), \binom{T}{2} - \Phi_{\min}(k, nT)] \neq \varnothing. \tag{3.6}$$

Proof. Equation (3.4) and (3.2) give the lower bound for T. Each n-secant contains $\binom{n}{2}$ pairs of points of $\mathcal{K}$, and each pair lies on one line, giving the upper bound. Apply Theorem 2.2 to the point–line design of the plane, using Lemma 3.1 for the degree bounds. □

If

$$v_{\mathcal{K}} \equiv nT \pmod{k}, \qquad 0 \leq v_{\mathcal{K}} < k, \qquad v_{\text{ext}} \equiv \tau T \pmod{b}, \qquad 0 \leq v_{\text{ext}} < b,$$

then the two minimum terms in (3.6) imply

$$T\left(\frac{n^2}{k} + \frac{\tau^2}{b} - 1\right) + \frac{v_{\mathcal{K}}(k - v_{\mathcal{K}})}{kT} + \frac{v_{\text{ext}}(b - v_{\text{ext}})}{bT} \leq q. \tag{3.7}$$

Discarding the two nonnegative residue terms gives the familiar real incidence bound. Since $T \geq \lambda b/\tau$, we obtain

$$\frac{\lambda b}{\tau}\left(\frac{n^2}{k} + \frac{\tau^2}{b} - 1\right) \leq q. \tag{3.8}$$

After denominators are cleared, (3.8) is the quadratic in Bishnoi–Mattheus–Schillewaert [7, Theorem 8.1]. Thus the classical bound is the real relaxation of (3.6); the residue terms record part of the discarded integer information. Schillewaert [15] determines the equality cases of the corresponding bound for minimal multiple blocking sets.

Remark 3.3. For a complete (k, n)-arc one takes $\lambda = 1$. The stronger values of λ below mean only that every external point is on at least λ n-secants. This is the same multiplicity hypothesis used in the dual form of Bishnoi–Mattheus–Schillewaert [7, Theorem 8.1].

4 Rational equality parameters

We first determine when the real incidence bound can be tight while the limiting number of n-secants through a point of the arc is an integer. Suppose that

$$\frac{n}{q} \longrightarrow \alpha, \qquad \frac{k}{q^2} \longrightarrow \beta,$$

that equality holds asymptotically in (3.8), and that the limiting value of $\rho_n(P)$ is an integer $a > \lambda$. Equality in the incidence and coverage bounds gives

$$a = \frac{\lambda\alpha(1-\beta)}{(1-\alpha)\beta}, \qquad \lambda(\alpha - \beta)^2 = \beta(1-\alpha). \tag{4.1}$$

If $d = a - \lambda$, eliminating β gives

$$d^2\alpha^2 - d(d+1)\alpha + a = 0, \qquad (d+1)^2 - 4a = (a - \lambda - 1)^2 - 4\lambda. \tag{4.2}$$

Lemma 4.1 (Classification of the rational parameters)**.** *Fix an integer $\lambda \geq 1$. The triples $(\alpha, \beta, a) \in \mathbb{Q}^2 \times \mathbb{Z}$ satisfying* (4.1),

$$0 < \beta < \alpha < 1, \qquad a > \lambda,$$

are in bijection with ordered factorizations $\lambda = uv$ in positive integers. They are

$$d = u + v + 1, \qquad \alpha = \frac{u+1}{u+v+1}, \qquad \beta = \frac{u}{u+v+1}, \qquad a = (u+1)(v+1). \tag{4.3}$$

Proof. The last expression in (4.2) must be an integer square. Factoring the difference of squares gives two positive factors of 4λ with the same parity. Both factors are even; write them as $2u$ and $2v$. Then $uv = \lambda$, and substitution gives (4.3). Conversely, the displayed values satisfy (4.1). The inequalities on α and β make the factors positive, and their order distinguishes the two roots, so the correspondence is one-to-one. $\square$

We next control $T = \tau_n$ near one of these parameters. The argument is stated separately because it is also used in the modular refinements.

Lemma 4.2 (Number of n-secants near equality)**.** *Fix $u, v \geq 1$ and constants $C, K \geq 0$, and put $d = u + v + 1$. Suppose*

$$q = dm, \qquad n = (u+1)m + 1, \qquad k = udm^2 + c_m m + e_m,$$

where $|c_m| \leq C$ and $|e_m| \leq K$. If every external point is on at least uv n-secants and

$$F(T) := \Phi_{\min}(k, nT) + \Phi_{\min}(q^2 + q + 1 - k, vmT) - \binom{T}{2} \tag{4.4}$$

is nonpositive, then

$$T = ud(v+1)m + O_{u,v,C,K}(1). \tag{4.5}$$

Proof. Set

$$a = (u+1)(v+1), \qquad \lambda = uv, \qquad T_* = ud(v+1)m,$$

and let $b = q^2 + q + 1 - k$. Thus

$$b = d(v+1)m^2 + (d - c_m)m + 1 - e_m.$$

The external incidence count gives

$$T \geq T_0 := \left\lceil \frac{uvb}{vm} \right\rceil = T_* + r_m, \tag{4.6}$$

where $|r_m| \leq R_0$ for a constant $R_0 = R_0(u, v, C, K)$.

We first record a uniform bound at this left endpoint. Since

$$nT_0 - ak = O_{u,v,C,K}(m), \qquad 0 \leq vmT_0 - \lambda b = O_{u,v,C,K}(m),$$

the balanced internal degrees at T_0 lie in $\{a-1, a, a+1\}$, and the balanced external degrees lie in $\{\lambda, \lambda+1\}$, for all sufficiently large m. Formula (2.1) therefore gives

$$\Phi_{\min}(k, nT_0) = k\binom{a}{2} + O_{u,v,C,K}(m), \qquad \Phi_{\min}(b, vmT_0) = b\binom{\lambda}{2} + O_{u,v,C,K}(m).$$

After substituting $k = udm^2 + O(m)$, $b = d(v+1)m^2 + O(m)$, and $T_0 = T_* + O(1)$, the coefficient of m^2 in

$$k\binom{a}{2} + b\binom{\lambda}{2} - \binom{T_0}{2}$$

vanishes by the identity

$$ud\binom{a}{2} + d(v+1)\binom{\lambda}{2} = \frac{u^2d^2(v+1)^2}{2}.$$

Consequently there is a constant $A = A(u, v, C, K)$ such that

$$|F(T_0)| \le Am \tag{4.7}$$

for all sufficiently large m.

Write $\Delta F(T) = F(T+1) - F(T)$. Its exact value is

$$\Delta F(T) = \sum_{i=0}^{n-1} \left\lfloor \frac{nT+i}{k} \right\rfloor + \sum_{i=0}^{vm-1} \left\lfloor \frac{vmT+i}{b} \right\rfloor - T. \tag{4.8}$$

For every $T \ge T_0$, the first family of floors is at least $a-1$, and the second is at least λ. Hence, uniformly for $0 \le s \le vm$,

$$\Delta F(T_0 + s) \ge n(a-1) + vm\lambda - T_0 - s = vm - s + O_{u,v,C,K}(1). \tag{4.9}$$

Choose a fixed integer $S > 2A/v + 2$. Equations (4.7) and (4.9) show that $F(T_0 + s) > 0$ for $S \le s \le vm/2$, once m is sufficiently large. From $vm/2$ to vm, the same lower bound is bounded below by a constant, while the accumulated increase at $vm/2$ has order m^2. Thus

$$F(T_0 + s) > 0 \qquad (S \le s \le vm). \tag{4.10}$$

It remains to exclude a later return to nonpositive values. Using $\lfloor x \rfloor \ge x - 1$ in (4.8) gives

$$\Delta F(T) \ge T\left(\frac{n^2}{k} + \frac{v^2m^2}{b} - 1\right) - (n + vm). \tag{4.11}$$

Here

$$\frac{n^2}{k} + \frac{v^2m^2}{b} - 1 = \frac{1}{u(v+1)} + O(m^{-1}).$$

At $T = T_0 + vm$, the right side of (4.11), divided by m, tends to

$$\frac{ud(v+1)+v}{u(v+1)} - d = \frac{v}{u(v+1)} > 0.$$

The coefficient of T in (4.11) is also positive for all sufficiently large m. Therefore $\Delta F(T) > 0$ for every $T \ge T_0 + vm$. Together with (4.10), this shows that $F(T) \le 0$ forces $0 \le T - T_0 < S$. Since $T_0 - T_* = O_{u,v,C,K}(1)$, this proves (4.5). □

Proof of Theorem 1.2. Lemma 4.1 proves the classification. If $(k - udm^2)/m$ exceeds a fixed constant larger than $c_{\mathbb{Z}}$, the claimed bound is automatic. In the remaining range, Lemma 4.2 applies uniformly to the actual number of n-secants, because (3.6) implies $F(T) \le 0$. Hence

$$T = ud(v+1)m + h$$

for an integer h in a fixed finite set. Put $c_m = (k - udm^2)/m$. Substitution in the external coverage inequality $uvb \le vmT$ gives

$$k \ge udm^2 + L(h)m - O_{u,v}(1). \tag{4.12}$$

At these parameters the balanced internal degrees are $(u+1)(v+1)-1$ and $(u+1)(v+1)$; the balanced external degrees are uv and $uv+1$. Substitution in (2.1) gives

$$\frac{F(T)}{m} = \frac{(u+v)(2uv+u+v+1)}{2}(C(h) - c_m) + O_{u,v}(m^{-1}). \tag{4.13}$$

The integer condition (3.6) implies $F(T) \le 0$, so

$$k \ge udm^2 + C(h)m - O_{u,v}(1).$$

The affine functions L and C are decreasing and increasing, respectively, and meet at h_*. Their maximum over the integers is minimized at $\lfloor h_* \rfloor$ or $\lceil h_* \rceil$. Since h ranges over a fixed finite set, the bounded terms are uniform. This proves (1.5).

Expanding the real bound (3.8) gives c_{mix}. Subtraction from c_{bal} gives the factorization (1.6). $\square$

For the first three ordered factorizations, the integer coefficients are

$$(u,v) = (1,1): \ \frac{18}{5}, \qquad (u,v) = (1,2): \ \frac{9}{2}, \qquad (u,v) = (2,1): \ 6. \tag{4.14}$$

The first is the ordinary-completeness family $q = 3m$, $n = 2m+1$. The other two have two n-secants through every external point and limiting ratios $n/q = 1/2$ and $3/4$.

There is also an elementary description of the parameters for which the coverage and pair-count inequalities are simultaneously tight to first order.

Proposition 4.3 (Simultaneous first-order equality)**.** *Put $M_0 = 2uv + u + v$. Then h_* is an integer if and only if*

$$M_0 \mid u^2(u+1)(u^2+u+1). \tag{4.15}$$

In particular, the case $v = u$ has simultaneous first-order equality for every even u.

Proof. Since $M_0 = (2u+1)v + u$, $\gcd(M_0, 2u+1) = \gcd(u, 2u+1) = 1$. Modulo M_0,

$$(2u+1)(v+1) \equiv u+1, \qquad (2u+1)(u^2v + u^2 - v) \equiv u(u^2+u+1).$$

It follows that

$$(2u+1)^2 u(v+1)(u^2v + u^2 - v) \equiv u^2(u+1)(u^2+u+1) \pmod{M_0}.$$

Because $2u+1$ is invertible modulo M_0, integrality of h_* is equivalent to (4.15). When $v = u$,

$$h_* = -\frac{u(u^2+u-1)}{2},$$

which is integral for even u. $\square$

5 Modular stability and completeness

Assume now that the plane is $\mathrm{PG}(2,q)$, where $q = p^e$. Dualizing the set of n-secants gives a point set $\mathcal{L}^*$ in the dual plane. The intersection of $\mathcal{L}^*$ with the line dual to a point X has size $\rho_n(X)$ or $\sigma_n(X)$, according as $X \in \mathcal{K}$ or $X \notin \mathcal{K}$. At a rational equality parameter from Section 4, the two limiting intersection numbers are

$$\lambda = uv, \qquad a = (u+1)(v+1), \qquad a - \lambda = u + v + 1.$$

When $u + v + 1$ is a power of p, these two numbers are congruent modulo p.

Szőnyi and Weiner prove that, for $q = p^e > 27$ with $e > 1$, a multiset in $\mathrm{PG}(2,q)$ having δ lines with the wrong intersection residue can, under their stated upper bound on δ, be changed at exactly $\lceil \delta/(q+1) \rceil$ distinct support points into a multiset having the prescribed residue on every line [16, Theorem 1.2]. In our application $e > 2$, $\delta = O(q)$, and their threshold is $(\lfloor \sqrt{q} \rfloor + 1)(q + 1 - \lfloor \sqrt{q} \rfloor)$. Thus the threshold holds for all sufficiently large q, and the number of changed support points is bounded. Related applications of modular stability to arc and direction problems appear in Csajbók–Weiner [9] and Adriaensen–Szőnyi–Weiner [2].

The following observation is where completeness re-enters the modular argument.

Lemma 5.1 (Contribution of the repair support)**.** *Fix an ordered factorization $\lambda = uv$ and suppose $d = u + v + 1$ is a power of the characteristic. Let*

$$q = dm, \qquad n = (u+1)m+1, \qquad k = udm^2 + cm + o(m), \qquad T = ud(v+1)m + h + o(1),$$

where $h \in \mathbb{Z}$. Suppose every external point is on at least λ n-secants, and suppose $\mathcal{L}^*$ can be changed at r distinct support points into a multiset meeting every line in $\lambda \pmod p$ points. If r is fixed and nonzero, then

$$c \geq L(h) + \frac{r}{u}. \tag{5.1}$$

Proof. Dualize the r support points to r lines of the original plane. Away from their pairwise intersections, a point on one of these lines has n-secant multiplicity different from $\lambda \pmod p$. If it is external, its multiplicity is at least $\lambda + 1$. A support line contains at most n points of $\mathcal{K}$, and the union loses only $O(r^2)$ points at pairwise intersections. Hence the total external excess over multiplicity λ is at least

$$r(q + 1 - n) - O(r^2) = rvm - O(r^2). \tag{5.2}$$

On the other hand, the incidence sum gives the exact first-order expression

$$(q + 1 - n)T - \lambda(q^2 + q + 1 - k) = uv(c - L(h))m + o(m). \tag{5.3}$$

Comparing (5.2) and (5.3) proves (5.1). □

There is a useful restriction when $r = 1$. Summing the line intersections of an exact $\lambda \pmod p$ multiset over a pencil shows that its total multiplicity is $\lambda \pmod p$. Along a characteristic-compatible family the leading term of T is divisible by p, so the correction has total multiplicity $h - \lambda \pmod p$. Consequently, if

$$h \equiv \lambda \pmod p, \tag{5.4}$$

then a nonzero correction cannot have support one.

We apply this first to ordinary completeness. Write $q = 3m$, $n = 2m + 1$, and suppose

$$k = 3m^2 + cm + o(m), \qquad T = 6m + h + o(1). \tag{5.5}$$

The coverage and pair-count inequalities from Sections 3 and 4 become

$$c + h \geq 3, \qquad 5c - h \geq 18. \tag{5.6}$$

Lemma 5.2 (The uncorrected case is impossible)**.** *Let $\mathcal{K}$ be a complete (k, n)-arc with parameters* (5.5)*, and let $\mathcal{L}^*$ be the dual point set of all its n-secants. If $c < 4$ and every line meets $\mathcal{L}^*$ in* $1 \pmod 3$ *points, then the exact pair identity* (3.3) *cannot hold.*

Proof. Near this equality parameter the balanced internal multiplicities are $3, 4$, and the balanced external multiplicities are $1, 2$. Let D be the signed internal deficiency $4k - nT$, and let E be the total external excess above one. The incidence sums give

$$D = (4c - 2h - 6)m + o(m), \tag{5.7}$$
$$E = (c + h - 3)m + o(m). \tag{5.8}$$

Completeness gives $E \geq 0$. Moreover, (5.6) and $c < 4$ give $D \geq (30 - 6c)m + o(m) > 0$. Without the congruence restriction, the available pair-count slack is

$$(5c - h - 18)m + o(m). \tag{5.9}$$

If every multiplicity is $1 \pmod 3$, congruence-restricted integer convexity replaces the internal adjacent values $3, 4$ by $1, 4$, raising the minimum pair count by D. It replaces the external adjacent values $1, 2$ by $1, 4$, raising that minimum by E. Feasibility would therefore require

$$5c - h - 18 \geq \frac{D + E}{m} + o(1) = 5c - h - 9 + o(1),$$

which is impossible. $\square$

Proof of Theorem 1.1. Suppose the assertion fails. For some $\varepsilon > 0$, pass to a sequence with

$$k \leq 3m^2 + (4 - 3\varepsilon)m.$$

The classical incidence bound supplies a lower bound for $c_m = (k - 3m^2)/m$, so, after passing to a subsequence, $c_m \to c$ with $c \leq 4 - 3\varepsilon$. The leading real inequality has a unique equality point at $T/m = 6$. Lemma 4.2 then gives $T - 6m = O(1)$; pass to a further subsequence on which this difference is a fixed integer h.

For the actual degree sequences, the pair identity gives the exact decomposition

$$-F(T) = \left(\sum_{P \in \mathcal{K}} \binom{\rho_n(P)}{2} - \Phi_{\min}(k, nT) \right) + \left(\sum_{Q \notin \mathcal{K}} \binom{\sigma_n(Q)}{2} - \Phi_{\min}(b, (q + 1 - n)T) \right).$$

Both parentheses are nonnegative. Equation (4.13), with bounded c_m and fixed h, shows that their sum is $O(m)$. Equation (2.3) therefore shows that only $O(m)$ internal degrees lie outside $\{3, 4\}$, and only $O(m)$ external degrees lie outside $\{1, 2\}$.

It remains to count the wrong adjacent degrees. The signed deficiency $D = 4k - nT = O(m)$ equals $\sum_{P \in \mathcal{K}} (4 - \rho_n(P))$. For every nonnegative integer $z \notin \{3, 4\}$,

$$|4 - z| \leq (z - 3)(z - 4),$$

so the total contribution of the nonadjacent internal degrees is $O(m)$ by (2.3). Hence the number of internal degree-three points is $O(m)$. Externally, completeness gives $\sigma_n(Q) \geq 1$, and the excess $E = \sum_Q (\sigma_n(Q) - 1) = O(m)$ bounds the number of degree-two points. Thus only $O(m) = O(q)$ dual lines fail to meet $\mathcal{L}^*$ in 1 (mod 3) points. Szőnyi–Weiner's repair theorem changes $\mathcal{L}^*$ at a bounded integer number r of support points to obtain an exact 1 (mod 3) multiset. Pass to a further subsequence on which r is fixed. Lemma 5.2 excludes $r = 0$. Lemma 5.1 gives

$$c \geq L(h) + r = 3 - h + r.$$

By (5.4), $r \geq 2$ when $h \equiv 1 \pmod 3$, and $r \geq 1$ otherwise. Together with the pair-count bound $c \geq (18 + h)/5$, this yields

$$c \geq \min_{h \in \mathbb{Z}} \max \left\{ \frac{18 + h}{5}, 3 - h + r_{\min}(h) \right\} = 4, \tag{5.10}$$

where $r_{\min}(h) = 2$ for $h \equiv 1 \pmod 3$ and is one otherwise. This contradicts the chosen sequence and proves (1.1). □

6 The characteristic-two family

We turn to the ordered factorization $(u, v) = (2, 1)$. Thus

$$q = 8m, \qquad n = 6m + 1, \qquad \lambda = 2.$$

Near equality write

$$k = 32m^2 + cm + o(m), \qquad T = 32m + h + o(1). \tag{6.1}$$

The integer pair condition and the external incidence count give

$$6c - h \geq 76, \qquad e = 2(c + h - 8)m + o(m), \tag{6.2}$$

where $e = \sum_{Q \notin \mathcal{K}} (\sigma_n(Q) - 2)$ is the total excess above the required two n-secants.

The dual set $\mathcal{L}^*$ has only $O(q)$ odd-secants. For sufficiently large even q, this is below the Szőnyi–Weiner threshold $(\lfloor \sqrt{q} \rfloor + 1)(q + 1 - \lfloor \sqrt{q} \rfloor)$. Their stability theorem [17, Theorem 1.1] therefore writes it as the symmetric difference

$$\mathcal{L}^* = \mathcal{E} \mathbin{\triangle} R, \tag{6.3}$$

where every line meets $\mathcal{E}$ in an even number of points and $|R| = r = \lceil \delta/(q + 1) \rceil = O(1)$, with δ the number of odd-secants of $\mathcal{L}^*$.

Lemma 6.1 (The even-type case). *Let $\mathcal{K}$ be a (k, n)-arc with parameters* (6.1)*, suppose every external point lies on at least two n-secants, and let $\mathcal{L}^*$ be the dual point set of all its n-secants. If $R = \varnothing$ in* (6.3)*, then $c \geq 16$.*

Proof. The balanced internal multiplicities are five and six, and the balanced external multiplicity is two. Put $D = 6k - nT$, the signed internal deficiency from six. The incidence equations and the unrestricted integer pair bound give

$$\begin{aligned} D &= (6c - 6h - 32)m + o(m), \\ e &= 2(c + h - 8)m + o(m), \\ \text{available pair slack} &= (12c - 2h - 152)m + o(m). \end{aligned}$$

If every multiplicity is even, restricting the internal balanced distribution to even integers raises its minimum by $|D|/2$, while the external restriction raises its minimum by $e/2$. The total increase is therefore at least

$$\frac{D+e}{2} = (4c - 2h - 24)m + o(m).$$

The available slack must be at least this large, which simplifies to $c \geq 16$. □

Lemma 6.2 (Parity of the repair)**.** *Suppose $R \neq \varnothing$ in* (6.3)*. If r is odd, then*

$$c + h \geq 9. \tag{6.4}$$

If r is even, then

$$c + h \geq 10. \tag{6.5}$$

Moreover, $r \equiv T \equiv h \pmod 2$.

Proof. Every even-type set has even cardinality, so $r \equiv |\mathcal{L}^*| = T \pmod 2$; the leading term $32m$ is even, giving the last congruence.

Dualize the points of R to r lines of the original plane. On each such line, at most $r-1$ points can cancel with the other generators in the symmetric difference. Thus at least $q - r + 2$ points have odd n-secant multiplicity. At most $n = 3q/4 + 1$ of them lie in $\mathcal{K}$, so a single generator contributes at least $q/4 - r + 1$ external points to the excess e.

If r is odd, then $r \geq 1$ and hence $e \geq q/4 - O(1) = 2m - O(1)$. If r is even, then $r \geq 2$; taking the union of two generator lines loses only their intersection, and $e \geq q/2 - O(1) = 4m - O(1)$. Comparing with (6.2) proves (6.4) and (6.5). □

Proof of Theorem 1.3. Suppose the theorem fails. Pass to a sequence for which the coefficient $c_m = (k - 32m^2)/m$ is bounded above by $73/6 - 8\varepsilon$. The classical incidence bound bounds c_m below, so pass to a subsequence with $c_m \to c$. The leading incidence bound and Lemma 4.2 give $T - 32m = O(1)$. Pass to a further subsequence with fixed integer difference h.

As in the characteristic-three proof, the exact decomposition of $-F(T)$ into the two nonnegative balancing losses and Equation (2.3) show that only $O(m)$ internal degrees lie outside $\{5, 6\}$, and only $O(m)$ external degrees lie outside $\{2, 3\}$. Moreover,

$$D = 6k - nT = \sum_{P \in \mathcal{K}} (6 - \rho_n(P)) = O(m).$$

For $z \notin \{5, 6\}$, the absolute value of $6 - z$ is at most twice the corresponding summand in (2.3); hence the number of internal degree-five points is $O(m)$. Since every external degree is at least two and $e = \sum_Q (\sigma_n(Q) - 2) = O(m)$, the number of external degree-three points is also $O(m)$. These are the wrong-parity adjacent degrees, so only $O(m) = O(q)$ dual lines have the wrong parity. Therefore Szőnyi–Weiner's theorem gives (6.3) with bounded integer r. Pass to a further subsequence on which r is fixed. Lemma 6.1 excludes $r = 0$ in the range under consideration. If h is odd, Lemma 6.2 and (6.2) give

$$c \geq \max\left\{\frac{76+h}{6}, 9 - h\right\}.$$

The minimum over odd h is $73/6$, attained at $h = -3$. If h is even, the corresponding bound is

$$c \geq \max\left\{\frac{76+h}{6}, 10 - h\right\},$$

whose minimum over even h is $37/3$, attained at $h = -2$. Both contradict the chosen sequence. Since $q = 8m$, this proves (1.7). □

The parity argument uses only the repair theorem and the fact that an n-secant contains at most n points of the arc. It does not require a classification of even-type sets of size $4q + O(1)$.

7 Blocking sets and projective codes

The complement of a (k, n)-arc $\mathcal{K}$ in a projective plane of order q is a $(q + 1 - n)$-fold blocking set. Completeness says more: each point of the complement lies on a line meeting it in exactly $q + 1 - n$ points. Thus the complement is a minimal multiple blocking set. Under this correspondence, n-secants of $\mathcal{K}$ are the $(q + 1 - n)$-secants of its complement, and the hypothesis $\sigma_n(Q) \geq \lambda$ says that every point of the blocking set is on at least λ such secants. This is precisely the dual setting of Bishnoi–Mattheus–Schillewaert [7, Section 8].

In particular, Theorem 1.1 is equivalent to the following statement.

Corollary 7.1. *Let $q = 3^h$ and let B be the complement of a complete $(k, 2q/3 + 1)$-arc in* $\mathrm{PG}(2, q)$. *Then B is a minimal $q/3$-fold blocking set and*

$$|B| \leq \frac{2q^2}{3} - \frac{q}{3} + 1 + o(q).$$

There is also a standard coding interpretation; compare Kohnert's incidence criterion for projective code extension [11] and the multiple-blocking-set/code correspondence of Ball–Fancsali [3]. If the (k, n)-arc spans the plane, projective representatives of its points form the columns of a generator matrix for a projective $[k, 3, k - n]_q$ code. A line meeting the arc in n points corresponds to a minimum-weight projective codeword. Adding a new projective column raises the minimum distance by one exactly when no minimum-weight hyperplane contains that column. Hence completeness is equivalent to nonextendibility. The condition $\sigma_n(Q) \geq \lambda$ records at least λ minimum-weight hyperplanes through every candidate column.

The same statement is available in every projective dimension $r - 1 \geq 2$. Points and hyperplanes of $\mathrm{PG}(r - 1, q)$ form a symmetric design with parameters

$$v = \frac{q^r - 1}{q - 1}, \qquad K = \frac{q^{r-1} - 1}{q - 1}, \qquad \Lambda = \frac{q^{r-2} - 1}{q - 1}. \tag{7.1}$$

Corollary 7.2 (Projective codes)**.** *Let C be a projective $[k, r, k - n]_q$ code, and let $\mathcal{S}$ be the hyperplanes containing n columns of a projective system for C. Partition the points of* $\mathrm{PG}(r - 1, q)$ *into the k columns of C and the remaining candidate columns. For every integer $T = |\mathcal{S}|$, the two selected-hyperplane degree sequences satisfy the exact interval condition* (2.4)*, with the parameters in* (7.1)*. Any prescribed lower bound on the number of hyperplanes in $\mathcal{S}$ through a candidate column may be used as the lower endpoint of its integer degree interval.*

Proof. Apply Theorem 2.2 to the point–hyperplane design of $\mathrm{PG}(r - 1, q)$. A minimum-weight hyperplane contains exactly n columns, so the selected blocks have constant intersection size n with the first point class. □

Exact two-valued hyperplane intersections correspond to projective two-weight codes [8]. Consider a near-equality sequence satisfying the hypotheses of Theorem 1.2, with bounded first-order coefficient and with the coverage and pair-count bounds simultaneously tight to first order at an integral h_* from Proposition 4.3. The balancing-loss argument in the proofs of Theorems 1.1 and 1.3 then shows that all but $O(q)$ dual lines have the two intersection numbers

$$uv \quad \text{and} \quad (u+1)(v+1).$$

Their difference is $u+v+1$, which must be a power of the characteristic along an infinite Desarguesian family. This agrees with the classical divisibility condition for exact projective two-weight codes. The present argument does not classify the $O(q)$ exceptional lines.

The exact two-character case does, however, admit a canonical inverse. The following observation is the dual incidence form of the classical two-character-set dictionary [8].

Proposition 7.3 (Exact two-character inverse)**.** *Let B be a nonempty proper point set in the dual of $\mathrm{PG}(2,q)$, and suppose both line characters $\lambda < a$ occur, with $\lambda \geq 1$. Let $\mathcal{S}$ be the primal lines dual to the points of B, put $d = a - \lambda$, and define*

$$\mathcal{A} = \{P : \deg_{\mathcal{S}}(P) = a\}.$$

Then $d \mid q$, and $\mathcal{A}$ is a complete (k,n)-arc whose full set of n-secants is $\mathcal{S}$, where

$$k = \frac{|B|(q+1) - \lambda(q^2+q+1)}{d}, \qquad n = \frac{|B| + q - \lambda(q+1)}{d}.$$

Every external point of $\mathcal{A}$ lies on exactly λ n-secants, and every line outside $\mathcal{S}$ meets $\mathcal{A}$ in $n - q/d$ points.

Proof. For a dual point x, let $N_a(x)$ be the number of lines through x that meet B in a points. Summing $|L \cap B|$ over the dual lines through x gives $|B| + q$ when $x \in B$, because x is counted on all $q+1$ lines, and gives $|B|$ when $x \notin B$. Therefore

$$N_a(x) = \begin{cases} (|B| + q - \lambda(q+1))/d, & x \in B, \\ (|B| - \lambda(q+1))/d, & x \notin B. \end{cases}$$

The two values differ by q/d, so $d \mid q$. Under duality, $N_a(\ell^*) = |\ell \cap \mathcal{A}|$. Thus the lines in $\mathcal{S}$ meet $\mathcal{A}$ in n points and every other line meets it in $n - q/d < n$ points. Hence $\mathcal{S}$ is exactly the full set of maximal secants. An external point has $\mathcal{S}$-degree λ, so $\lambda \geq 1$ proves completeness and the stated multiplicity.

Finally, summing $|L \cap B|$ over all dual lines gives

$$|B|(q+1) = ak + \lambda(q^2+q+1-k),$$

which yields the formula for k. $\square$

8 Concluding remarks

The first two incidence moments already contain the classical spectral bound. Their sharp integer degree ranges retain information that the real relaxation loses, and at each rational equality parameter this loss has linear order. Modular stability then converts the nearly two-valued dual intersection distribution into a bounded repair. Completeness assigns a further linear cost to every point in that repair support.

Proposition 7.3 closes the inverse-realization problem for an exact two-character dual set: the high-character lines recover the arc canonically, and the dual point set is exactly its maximal-secant family. The remaining inverse problem begins only after bounded repair, where the perturbed modular core need not retain exact maximal intersections.

The characteristic-three result leaves a concrete extremal problem:

$$t_{2q/3+1}(2,q) \stackrel{?}{=} \frac{q^2}{3} + \frac{4q}{3} + o(q) \qquad (q = 3^h \to \infty). \tag{8.1}$$

A matching construction would determine the first two asymptotic terms. A second problem is structural: classify the sets of dual points with only $O(q)$ lines outside the two predicted intersection numbers and determine which of them can occur as the full set of n-secants of a complete (k,n)-arc.

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